

Exercise 2.17. Assume that a is an integer and p and q are distinct primes. Prove that if $p \mid a$ and $q \mid a$, then $pq \mid a$.

Solution.

We are given:

- a is an integer.
- p and q are distinct primes.
- $p \mid a$ and $q \mid a$.

We want to prove:

$$\boxed{pq \mid a}.$$

Proof

Since

$$p \mid a,$$

by the definition of divisibility, there exists an integer k such that

$$a = pk.$$

Also,

$$q \mid a,$$

so

$$q \mid pk.$$

Now p and q are distinct primes, so

$$p \neq q.$$

Because q is prime, its only positive divisors are 1 and q . Since $p \neq q$,

$$q \nmid p.$$

By **Euclid's Lemma**, if a prime q divides a product pk , then q must divide one of the factors:

$$q \mid p \quad \text{or} \quad q \mid k.$$

But $q \nmid p$, so necessarily

$$q \mid k.$$

Therefore, there exists an integer m such that

$$k = qm.$$

Substitute this into $a = pk$:

$$a = p(qm).$$

Thus,

$$a = pqm.$$

Since m is an integer,

$$\boxed{pq \mid a}.$$

Short version

$$p \mid a \Rightarrow a = pk$$

and $q \mid a$ gives

$$q \mid pk.$$

Since p, q are distinct primes, Euclid's Lemma gives

$$q \mid k.$$

Thus $k = qm$, so

$$a = pk = p(qm) = pqm.$$

Hence

$$\boxed{pq \mid a}.$$